

Interpretable and Efficient Penalized Multivariate Monomial Splines

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Abstract

We propose a computationally efficient method for fitting penalized piecewise polynomial splines directly in monomial form, where smoothness at partition boundaries is enforced through explicit linear constraints on the coefficients rather than construction of a smooth basis. The constrained penalized estimate is obtained via a transformed ordinary least squares projection that exploits block-diagonal structure of the design matrix across partitions, avoiding large dense systems and enabling parallel computation, which is preserved for extensions to generalized linear models, shape constraints, and certain covariance structures. A closed-form integrated curvature penalty analogous to cubic smoothing splines is used for regularization. In simulations, the proposed method gives competitive accuracy with B-spline-based smoothers while often being faster, especially at larger sample sizes, and retaining a local polynomial representation that is directly interpretable. In a pharmacokinetic application, the monomial framework supports simultaneous shape constraints, structured within-subject correlation, and direct extraction of non-compartmental parameter summaries with fully propagated posterior uncertainty in a single fit. The practical advantages include computational efficiency relative to common spline software, and a fitted function that remains interpretable as local polynomial equations. Implementation is available in the `lgspline` package for R.

Keywords: Bayesian polynomial splines; Constrained optimization; Interpretability; Parallelism; Shape-constrained splines; Smoothing spline penalty.

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1 Introduction

Polynomial splines are widely used for modeling nonlinear predictor-response relationships across science and mathematics. Modern spline methodology is largely built around basis constructions that enforce smoothness implicitly, most notably B-splines (Cox, 1972; de Boor, 1972). These representations are numerically stable and computationally efficient, and are the foundation of most widely used software. Truncated power bases can span the same function space and are more straightforward to implement without specialized software for B-spline bases, but are less computationally stable (Stone and Koo, 1985; Ezhov, Neitzel, and Petrovic, 2021). Related approaches for fitting splines include M- and I-splines (Schumaker, 2007; Ruppert, Wand, and Carroll, 2003; Ramsay, 1988), thin-plate splines (Duchon, 1977; Wood, 2003), smoothing spline ANOVA (Wahba, 1990; Gu, 2002), and many others. Chapters 4 and 5 of Wood (2017) may be referenced for an introductory history and overview of spline construction in the context of generalized additive models (GAMs), and Ezhov, Neitzel, and Petrovic (2021) provides the explicit relationship between monomial, truncated power, and B-spline bases.

For several spline bases, most prominently B-splines and truncated power bases, the fitted function is a sum of piecewise polynomials and can in principle be rearranged into a monomial basis (Ezhov, Neitzel, and Petrovic, 2021). In standard implementations, however, this structure is implicit and not always accessible. Recovering explicit polynomial expressions requires nontrivial post-fitting algebraic rearrangement that is generally unavailable in modern software. This difficulty becomes substantial in multivariate settings with standardized predictors and many interaction terms from tensor product operations, leading to combinatorial growth in the complexity of the bookkeeping, and even more work to calculate statistics for inference, such as standard errors. For other constructions such as polyharmonic splines or kernel representations, rearrangement of the fitted function into polynomial form exactly is not possible.

Expressing the fitted model directly in monomial form, sometimes referred to as Vandermonde or power form, provides an immediately interpretable representation with natural extensions to higher dimensions through simple interaction terms. In this form, derivatives,

integrals, and extrema are straightforward to express, and the model can be communicated or evaluated without reference to knots, basis functions, or the method used to fit it.

In contrast, B-splines and truncated power splines are often considered difficult to interpret directly from the model output in terms of basis expansions and coefficients alone (Ezhov, Neitzel, and Petrovic, 2018; Acharjee and Das, 2022). While many statisticians are comfortable viewing splines through a purely nonparametric lens, a parametric interpretation is still desired in practice by subject-matter researchers, especially in biomedical research. Examples include applications to pharmacokinetic modeling (Yeh and Small, 1989; Popović, 2006), epidemiology (Schuster, Rijnhart, Twisk, and Heymans, 2022), exposure-response analyses (Malloy, Kapellusch, and Garg, 2017), and meta-analyses (Belias et al., 2022; Petropoulou, Rücker, and Schwarzer, 2026). For clinical trial submissions in particular, having an interpretable estimand expressible as a linear combination of one or several coefficients from a parametric model targeting a predictor-response relationship, and that can be written out cleanly with notation for a statistical analysis plan, is not optional for many regulatory bodies. Statistical models for safety and pharmacokinetic endpoints have to be interpretable by non-statisticians in real-time during meetings, often with the statistician who fit the model absent for blinding purposes.

An alternative formulation of splines as piecewise polynomial regression models subject to linear restrictions on the coefficients enforcing smoothness was explored early on by Buse and Lim (1977) and Fuller (1969). Ezhov, Neitzel, and Petrovic (2018) showed that the linear restrictions can be expressed as external constraints and the spline can be fit directly in monomial form using the method of Lagrangian multipliers. They do not take credit for the method, suggesting it was so intuitive that it had likely been rediscovered several times over, although they could not find a previous reference. A similar constrained monomial formulation was independently developed by the author (Davis, 2023) as a component of a hyperparameter optimization algorithm. The formulation was not the focus of that work, nor thought to be original. The approach has otherwise seen limited adoption, perhaps because the spline basis in the single-predictor case is a poorly conditioned Vandermonde matrix, the approach requires the additional step of constructing external constraints,

and the resulting system of equations is much greater in dimension than for B-splines or truncated power splines when fit naïvely. To the best of our knowledge, the monomial formulation has never been formally explored with multiple predictors, regularization, nor outside of the scenario of Gaussian-distributed independent homoscedastic error terms.

In this article we develop a computational framework that revisits the constrained formulation and makes direct fitting in monomial form computationally practical. The main idea is to exploit the block-diagonal structure of the model’s design matrix, and compute the constrained estimate by solving a transformed ordinary least squares (OLS) problem that depends only on the number of constraints, rather than the number of coefficients directly. This avoids construction of large dense matrices and enables parallel computation for most model-fitting operations. The procedure extends straightforwardly to generalized linear models (GLMs), shape constraints, and for incorporating certain covariance structures.

The specific contributions are as follows. First, we develop a constrained penalized estimation framework in which smoothness is enforced through explicit linear constraints. Second, we derive a transformed OLS projection that exploits block-diagonal structure for computational and memory efficiency. Third, we establish extensions that preserve this structure for non-Gaussian response, inequality constraints including shape constraints, and sparse covariance structures. Fourth, we provide a closed-form curvature penalty defined on the monomial basis similar to cubic smoothing spline penalties. Fifth, we present empirical studies demonstrating advantages in predictive performance, computational scaling, and practical interpretability.

The remainder of the paper is organized as follows. Section 2 covers the model formulation, the computational strategy, extensions for generalized linear models and structured covariance, Bayesian inference under constraint, and practical considerations. Section 3 presents empirical results, beginning with a simulation study, followed by a pharmacokinetic application and a parallelism benchmark. Section 4 concludes.

2 Materials and Methods

2.1 Piecewise Polynomial Structure

Consider data consisting of N observations with q -dimensional predictor vectors $\mathbf{t}_i \in \mathbb{R}^q$ and scalar responses $y_i \in \mathbb{R}$, let f be some unknown but smooth function $f : \mathbb{R}^q \rightarrow \mathbb{R}$, and suppose that $y_i \mid \mathbf{t}_i, \sigma^2 \sim \mathcal{N}(f(\mathbf{t}_i), \sigma^2)$ with $\text{Cov}(y_i, y_{i'}) = 0$ for $i \neq i'$. We wish to estimate f .

First, we partition the predictor space into $K+1$ mutually exclusive regions $\mathcal{P}_0, \dots, \mathcal{P}_K$, so that each observation belongs to exactly one partition. Within partition k , we approximate f with a polynomial:

$$\hat{f}_k(\mathbf{t}) = \mathbf{x}(\mathbf{t})^\top \boldsymbol{\beta}_k, \quad (1)$$

where $\mathbf{x}(\mathbf{t}) \in \mathbb{R}^p$ collects monomial basis terms (intercept, polynomial, and interaction terms as needed) and $\boldsymbol{\beta}_k \in \mathbb{R}^p$ is the coefficient vector for partition k . The $i = 1, \dots, n_k$ expansions $\mathbf{x}_i$ in partition k can be expressed as an $n_k \times p$ design matrix $\mathbf{X}_k = (\dots, \mathbf{x}_i, \dots)^\top$.

In the simple case of a single predictor and one partition with cubic expansions, $\mathbf{x}(t) = (1, t, t^2, t^3)^\top$ and the fitted function is $\hat{f}(t) = \hat{\beta}_0 + \hat{\beta}_1 t + \hat{\beta}_2 t^2 + \hat{\beta}_3 t^3$. Derivatives follow directly from the power rule, for example $\hat{f}'(t) = \mathbf{x}'(t)^\top \hat{\boldsymbol{\beta}} = \hat{\beta}_1 + 2\hat{\beta}_2 t + 3\hat{\beta}_3 t^2$. Unlike B-splines and truncated powers, for this formulation the expansions are the same in every partition and avoid reference of knot locations.

Let the full coefficient vector be $\boldsymbol{\beta} = (\boldsymbol{\beta}_0^\top, \dots, \boldsymbol{\beta}_K^\top)^\top \in \mathbb{R}^P$ with $P = p(K+1)$. The $N \times P$ design matrix is block-diagonal, $\mathbf{X} = \text{blockdiag}(\mathbf{X}_0, \dots, \mathbf{X}_K)$, with blocks of dimension $n_k \times p$, where the total sample size is $N = \sum_{k=0}^K n_k$.

Let $\boldsymbol{\Lambda} = \text{blockdiag}(\boldsymbol{\Lambda}_0, \dots, \boldsymbol{\Lambda}_K)$ be a block-diagonal penalty matrix of dimension $P \times P$ with positive-definite $p \times p$ blocks (see Section 2.3). The model can be fit using penalized least squares:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \frac{1}{2} \boldsymbol{\beta}^\top \boldsymbol{\Lambda} \boldsymbol{\beta} \right\}. \quad (2)$$

The solution is the usual penalized maximum likelihood (MLE) estimate, equivalently, the maximum *a posteriori* estimate, $\hat{\boldsymbol{\beta}} = \mathbf{G}\mathbf{X}^\top \mathbf{y}$, where $\mathbf{G} = (\mathbf{X}^\top \mathbf{X} + \boldsymbol{\Lambda})^{-1}$ is block-diagonal with assumed positive-definite $p \times p$ blocks $\mathbf{G}_k = (\mathbf{X}_k^\top \mathbf{X}_k + \boldsymbol{\Lambda}_k)^{-1}$. For each partition $\mathcal{P}_k$

the penalized estimate is simply $\hat{\beta}_k = \mathbf{G}_k \mathbf{X}_k^\top \mathbf{y}_k$, so fitting can proceed in parallel across partitions.

2.2 Smoothness Constraints

Without constraint the piecewise polynomial is discontinuous at partition boundaries. Smoothness is imposed by requiring the polynomial and its derivatives to agree at shared knot locations (Fuller, 1969; Buse and Lim, 1977; Ezhov, Neitzel, and Petrovic, 2018). At each knot ξ between neighboring partitions k and k^* , the function and its first $d - 1$ derivatives are constrained to match:

$$f_k(\xi) = f_{k^*}(\xi), \quad f'_k(\xi) = f'_{k^*}(\xi), \quad \dots, \quad f_k^{(d-1)}(\xi) = f_{k^*}^{(d-1)}(\xi). \quad (3)$$

For example, for a cubic spline with $d = 3$, this enforces continuity of the function, its first derivative, and its second derivative. With multiple predictors, the second derivative is the total second derivative (see Supplement A).

With multiple spline effects, an issue arises in that the fitted function can become over-constrained, leading to suboptimal performance at best, and collapse of the fitted function to a single polynomial identical across partitions at worst. One alternative proposed here is to instead constrain the *sum* of the derivative constraints to be equivalent, for example with $d = 3$:

$$f_k(\xi) = f_{k^*}(\xi), \quad f'_k(\xi) + f''_k(\xi) = f'_{k^*}(\xi) + f''_{k^*}(\xi). \quad (4)$$

This reduces the number of constraints per knot from $1 + (d - 1)q$ to $1 + (d - 2)q$. Since the expansions $\mathbf{x}'(\mathbf{t}), \mathbf{x}''(\mathbf{t})$ for the first derivative and second derivatives are linearly independent (linear terms $\mathbf{t}^1$ are always “1” for first derivatives, and always “0” for second derivatives), this is *likely* to result in both constraints being met. The technique is still a heuristic; it is possible the sum of derivative constraints can be met while individually violated.

In either construction, the constraint count is driven by the number of spline-expanded predictors and neighboring partitions. It does not increase when higher-order polynomial terms or interaction terms are added to the basis.

These constraints are linear in $\boldsymbol{\beta}$ (Fuller, 1969; Buse and Lim, 1977; Ezhov, Neitzel, and Petrovic, 2018) and can be collected into a matrix $\mathbf{A}$ of dimension $P \times R$ such that $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{0}$ implies the smoothness conditions are satisfied, illustrated in Supplement A. A pivoted QR decomposition of $\mathbf{A}$ improves computational stability and prevents redundant constraints in practice. The constraint matrix is typically underdetermined ($R < P$), because adding polynomial terms and interactions increases the model dimension without adding new smoothness conditions (see Supplement A.2). Columns can be added to $\mathbf{A}$ for user-supplied equality constraints, such as when fitting nested models, and specific smoothness conditions can be removed to relax the fit, a flexibility that is unique and central to the framework.

The constrained solution follows from Lagrangian multipliers (Appendix A). Define the $P \times P$ projection matrix

$$\mathbf{U} = \mathbf{I} - \mathbf{GA}(\mathbf{A}^\top \mathbf{GA})^{-1} \mathbf{A}^\top, \quad (5)$$

where $\mathbf{G} = \text{blockdiag}(\mathbf{G}_0, \dots, \mathbf{G}_K)$. The constrained estimate is

$$\tilde{\boldsymbol{\beta}} = \mathbf{U} \hat{\boldsymbol{\beta}}. \quad (6)$$

Fitted values are $\tilde{\mathbf{y}} = \mathbf{X} \tilde{\boldsymbol{\beta}} = \mathbf{H} \mathbf{y}$ for hat matrix $\mathbf{H} = \mathbf{X} \mathbf{U} \mathbf{G} \mathbf{X}^\top$. For non-zero constraints ($\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{A}^\top \boldsymbol{\beta}_0 = \mathbf{c}$), the solution becomes $\tilde{\boldsymbol{\beta}} = \mathbf{U} \hat{\boldsymbol{\beta}} + (\mathbf{I} - \mathbf{U}) \boldsymbol{\beta}_0$.

The matrix $\mathbf{U}$ is idempotent and satisfies $\mathbf{A}^\top \mathbf{U} = \mathbf{0}$ and $\mathbf{U} \mathbf{G} \mathbf{U}^\top = \mathbf{U} \mathbf{G}$. Effective degrees of freedom are $\nu = \text{tr}(\mathbf{H})$ and for Gaussian identity-link models a variance estimate $\tilde{\sigma}^2 = \|\mathbf{y} - \tilde{\mathbf{y}}\|^2 / (N - \nu)$ (see discussion of “Effective degrees of freedom” by Wood (2017)) can be obtained by maximizing the restricted maximum likelihood (REML) objective as described in Supplement C.

For Gaussian identity-link models, the exact frequentist variance-covariance matrix of $\tilde{\boldsymbol{\beta}}$ is

$$\widetilde{\text{Var}(\tilde{\boldsymbol{\beta}})} = \tilde{\sigma}^2 \mathbf{U} \mathbf{G} (\mathbf{X}^\top \mathbf{X}) \mathbf{G} \mathbf{U}^\top = \tilde{\sigma}^2 \mathbf{U} \mathbf{G} - \tilde{\sigma}^2 \mathbf{U} \mathbf{G} \boldsymbol{\Lambda} \mathbf{G} \mathbf{U}^\top, \quad (7)$$

using $\mathbf{G}(\mathbf{X}^\top \mathbf{X}) \mathbf{G} = \mathbf{G}(\mathbf{G}^{-1} - \boldsymbol{\Lambda}) \mathbf{G}$. The first term $\tilde{\sigma}^2 \mathbf{U} \mathbf{G}$ is the Bayesian posterior variance-covariance matrix under the interpretation of the penalty matrix as a precision matrix for normal priors on coefficients (see Section 2.6), and subtracting the second term

yields the frequentist exact estimate. For non-Gaussian models, `lgspline` instead uses the usual large-sample Laplace approximation based on the fitted Hessian. For example, for exponential family models, $\mathbf{G}$ is replaced with the inverse negative Hessian of the penalized log-likelihood scaled by the inverse of an exponential dispersion scalar, if present (see Section 2.5.2).

2.3 Integrated Curvature Penalty

Regularization for the proposed method uses essentially the equivalent of a cubic smoothing spline penalty with one predictor (Reinsch, 1967), and a modified thin-plate spline penalty with multiple predictors (Duchon, 1977; Wood, 2003), with a ridge penalty included for only intercepts and linear terms. Unlike simple ridge regression, this penalty allows higher polynomial terms to be penalized differently than lower-order polynomial terms in a meaningful way that does not rely on heuristics or additional hyperparameters. Theoretically, the penalty is closely related to the norm defining the topology of a Sobolev space $\mathcal{W}_2^2$ which Reinsch (1967) showed is minimized by some piecewise cubic function, thus justifying the univariate cubic smoothing spline. P-spline penalties (Eilers and Marx, 1996) are popular in practice and naturally suited for B-splines due to their banded structure, but by penalizing differences across partitions they cannot be expressed in block-diagonal form, and so are not as amenable to the proposed monomial formulation.

In its simplest form, the regularization scheme proposed for the monomial spline formulation involves constructing a quadratic penalty matrix $\mathbf{\Lambda}_k$ for each partition expressible as an elementwise sum of a cubic smoothing spline penalty $\mathbf{\Lambda}_s$ weighted by a “wobble penalty” $\lambda_w \geq 0$ and a “flat ridge penalty” for non-spline terms $\mathbf{\Lambda}_r$ weighted by $\lambda_r \lambda_w \geq 0$, such that $\mathbf{\Lambda}_k = \lambda_w(\mathbf{\Lambda}_s + \lambda_r \mathbf{\Lambda}_r)$. Additional predictor-specific and partition-specific penalties can be included as well:

$$\mathbf{\Lambda}_k = \lambda_w \left(\mathbf{\Lambda}_s + \lambda_r \mathbf{\Lambda}_r + \sum_{l=1}^L \lambda_{l,k} \mathbf{\Lambda}_{l,k} \right), \quad (8)$$

for additional penalties $\lambda_{l,k} > 0$ and penalty matrices $\mathbf{\Lambda}_{l,k}$ possibly unique to the k th partition or l th predictor. The full $P \times P$ penalty matrix $\mathbf{\Lambda} = \begin{pmatrix} \ddots & & \\ & \mathbf{\Lambda}_k & \\ & & \ddots \end{pmatrix}$ is block-

diagonal of $K + 1$, $p \times p$ blocks. The parameterization is chosen such that, once derivatives of λ_w are computed, the derivatives for λ_r and $\lambda_{l,k}$ are faster to compute for purposes of gradient-based optimization; details go beyond this paper but are documented in **lgspline**.

The penalty matrix $\mathbf{\Lambda}_s$ serves as the integrated curvature penalty matrix, and can be derived as follows for a single predictor with minimum a and maximum b observed in the dataset:

$$\int_a^b \|f''(t)\|^2 dt = \int_a^b \|\mathbf{x}''(t)^\top \boldsymbol{\beta}_k\|^2 dt = \boldsymbol{\beta}_k^\top \left\{ \int_a^b \mathbf{x}''(t) \mathbf{x}''(t)^\top dt \right\} \boldsymbol{\beta}_k = \boldsymbol{\beta}_k^\top \mathbf{\Lambda}_s \boldsymbol{\beta}_k. \quad (9)$$

For illustration in a single partition with cubic basis $\mathbf{x}(t) = (1, t, t^2, t^3)^\top$ and $\mathbf{x}''(t) = (0, 0, 2, 6t)^\top$:

$$\mathbf{\Lambda}_s = \int_a^b \begin{pmatrix} 0 \\ 0 \\ 2 \\ 6t \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 2 \\ 6t \end{pmatrix}^\top dt = \int_a^b \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 12t \\ 0 & 0 & 12t & 36t^2 \end{pmatrix} dt = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 4(b-a) & 6(b^2-a^2) \\ 0 & 0 & 6(b^2-a^2) & 12(b^3-a^3) \end{pmatrix}, \quad (10)$$

which is equivalent to the classical cubic smoothing spline penalty of Reinsch (1967) applied to a single partition.

For multiple predictors, an elementwise sum of this operation over all predictors is taken. The second derivative is computed as a total second derivative, accounting for interaction terms, such that for q predictors $\tilde{\boldsymbol{\beta}}_k^\top \mathbf{\Lambda}_s \tilde{\boldsymbol{\beta}}_k = \sum_{j=1}^q \int_{a_j}^{b_j} \left(\frac{\partial^2}{\partial t_j^2} \tilde{f}_k(\mathbf{t}) + \sum_{j^* \neq j} \frac{\partial^2}{\partial t_j \partial t_{j^*}} \tilde{f}_k(\mathbf{t}) \right)^2 dt_j$.

A ridge penalty matrix is included for computational stability of non-spline terms and helps impose positive definiteness of $\mathbf{\Lambda}$, since the second derivative of linear terms and intercepts is 0, so $\mathbf{\Lambda}_s$ is positive semidefinite. For the single-predictor cubic scenario:

$$\mathbf{\Lambda}_r = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (11)$$

Additional tunable penalties and matrices for partition- or predictor-specific penalization are straightforward to include, for example by isolating the elements of $\mathbf{\Lambda}_s$ that

involve only the j th predictor and up-weighting relative to the j^* th predictor, thus relaxing isotropic assumptions with negligible computational cost. By default, `lgspline` tunes penalties using a constrained version of leave-one-out cross-validation (Tarpey, 2000), with optional generalized cross-validation (Golub, Heath, and Wahba, 1979) and optional L^2 regularization on penalty terms that biases them toward 1 (thus a null multiplicative effect). Further details are given in the package documentation. REML is commonly used for tuning penalized splines (Wood, 2017) and is computationally attractive but was not implemented to avoid confusion with covariance parameter estimation here.

2.4 Relationship to Other Spline Representations

The practical distinction of the present framework is that the model is fit directly in monomial form. For illustration, in the cubic single-predictor case the fitted function across all partitions is $\tilde{f}(t) = \sum_{k=0}^K (\tilde{\beta}_{0,k} + \tilde{\beta}_{1,k}t + \tilde{\beta}_{2,k}t^2 + \tilde{\beta}_{3,k}t^3)\mathbb{I}(t \in \mathcal{P}_k)$, sharing the same functional form without reference to knot locations. For B-spline bases the fitted function is $\tilde{f}(t) = \sum_{j=1}^P B_j(t)\tilde{\beta}_j$, where the B_j are B-spline basis functions with compact support, which do not reveal the underlying predictor-response relationship without algebraic rearrangement. For truncated power bases the fitted function may be written as a polynomial plus a cumulative sum of polynomial expansions shifted by knot locations. For example, in the single-predictor cubic case $\tilde{f}(t) = \tilde{\beta}_0 + \tilde{\beta}_1t + \tilde{\beta}_2t^2 + \tilde{\beta}_3t^3 + \sum_{m=1}^K \tilde{\beta}_{m+3}(t - \xi_m)_+^3$, so the representation is explicit but accumulates shifted polynomial terms that directly reference knot locations unless it is rearranged within each partition. Unlike B-splines and the proposed formulation, truncated power bases do not offer compact support, and all coefficient estimates are needed to describe the predictor-response relationship in at least one partition. Each has its own advantage: the monomial parameterization used here fits the local polynomial directly and prioritizes interpretability; the B-spline representation favors numerical stability; and the truncated power representation is the simplest to implement without devoted software for imposing constraints or constructing B-spline bases.

2.5 Computation

2.5.1 Projection via Transformed OLS

The estimator (6) involves the $P \times P$ matrix $\mathbf{U}$, which is expensive to form when P is large. The central computational device of this article avoids constructing $\mathbf{U}$ entirely by exploiting the block-diagonal structure of $\mathbf{X}$, $\mathbf{G}$, and $\mathbf{\Lambda}$. The constrained estimator $\tilde{\boldsymbol{\beta}}$ is obtained as the residuals of an ordinary least-squares problem left-multiplied by $\mathbf{G}^{1/2}$ in four steps:

1. Compute the partition-wise unconstrained estimates $\hat{\boldsymbol{\beta}}$.
2. Transform: $\mathbf{y}^* = \mathbf{G}^{-1/2}\hat{\boldsymbol{\beta}}$, $\mathbf{X}^* = \mathbf{G}^{1/2}\mathbf{A}$.
3. Fit $\mathbb{E}[\mathbf{y}^*] = \mathbf{X}^*\boldsymbol{\gamma}$ for coefficients $\boldsymbol{\gamma}$ by OLS via QR decomposition.
4. Recover $\tilde{\boldsymbol{\beta}} = \mathbf{G}^{1/2}\mathbf{r}^*$ where $\mathbf{r}^* = \mathbf{y}^* - \mathbf{X}^*\hat{\boldsymbol{\gamma}}$.

Back-substitution confirms $\mathbf{G}^{1/2}\mathbf{r}^* = \mathbf{U}\hat{\boldsymbol{\beta}} = \tilde{\boldsymbol{\beta}}$ (Appendix A). Steps 1 and 2, the construction of $\mathbf{A}$, the multiplications by $\mathbf{X}$, and the eigendecomposition of $\mathbf{G}$ used for matrix square roots can all be computed in parallel across $K + 1$ partitions. The most expensive operation is the QR decomposition of the $P \times R$ matrix $\mathbf{X}^*$ in step 3. Since $R \leq P$ typically holds (see Supplement A), this is more efficient than the QR decomposition of the $(N + R) \times P$ matrix used for solving the constrained system described by Ezhov, Neitzel, and Petrovic (2018), and compares favorably to the banded operations of B-splines, which are not as straightforwardly parallelizable.

The matrices $\mathbf{G}$, $\mathbf{X}$, and $\mathbf{\Lambda}$ are also memory-efficient; in R, as implemented in `lgspline`, they are stored as a list of $K + 1$ small $p \times p$ matrices rather than collapsed into $P \times P$ block-diagonal form, an advantage that cannot be replicated with truncated power bases.

2.5.2 Extensions

GLMs. The same transformed-OLS framework extends naturally beyond the independent Gaussian setting. For GLM responses with working weights $w_i = [V(\mu_i)\{g'(\mu_i)\}^2]^{-1}$, the per-partition scaled inverse information matrix $\mathbf{G}_k$ is now a function of $\boldsymbol{\beta}$ and takes

the form $\mathbf{G}_k = (\mathbf{X}_k^\top \mathbf{W}_k \mathbf{X}_k + \mathbf{\Lambda}_k)^{-1}$ with $\mathbf{W}_k = \text{diag}(w_i : i \in \mathcal{P}_k)$. Because $\mathbf{W}$ is diagonal the block-diagonal structure is preserved, and the four-step procedure applies with the weighted $\mathbf{G}_k$. Although the estimator is not derived as the unique constrained penalized MLE, it is consistent and asymptotically converges to a normal distribution under standard regularity conditions (Supplement D).

When the link is non-identity, $\tilde{\mathbf{G}} = \mathbf{G}(\tilde{\boldsymbol{\beta}})$ depends on fitted values through the working weights. At the m th iteration, the approach holds $\hat{\boldsymbol{\beta}}$ fixed at the partition-wise unconstrained solution, recomputes $\mathbf{G}^{(m)}$ using the constrained estimate $\tilde{\boldsymbol{\beta}}^{(m)}$, and re-applies the projection: $\tilde{\boldsymbol{\beta}}^{(m+1)} = \tilde{\mathbf{U}}^{(m)} \hat{\boldsymbol{\beta}}$, iterating until the mean absolute coefficient change falls below a tolerance without worsening in-sample performance. Convergence is typically rapid, usually fewer than 10 iterations. In practice, the computational technique of Section 2.5.1 can be used to avoid explicit construction of $\tilde{\mathbf{U}}^{(m)}$ for each iteration. Note that the unconstrained estimates only need to be computed once. The supplement `simple_illustration.R` provides code for demonstrating this with spline logistic regression.

Shape Constraints. For C general inequality constraints $\mathbf{C}^\top \boldsymbol{\beta} \succeq \mathbf{c}$ with conformable $\mathbf{C} \in \mathbb{R}^{P \times C}$, $\mathbf{c} \in \mathbb{R}^C$ including shape constraints, the constraints enter through the matrix already present. This is achieved using an active-set method, whereby the inequality constraints that happen to be met using the most recent estimate of $\boldsymbol{\beta}$ are ignored as “inactive”, while inequality constraints that are not yet met get appended to $\mathbf{A}$ as “active” equalities, $\mathbf{A}_{\text{aug}} = [\mathbf{A} \mid \mathbf{C}_A]$, and the same four-step transformed OLS remains unchanged after. The active set is updated by checking primal and dual feasibility until the Karush-Kuhn-Tucker conditions hold (Nocedal and Wright, 2006).

Correlated Response. With correlated response and working covariance matrix $\mathbf{V}$, $\mathbf{G}^{-1} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} + \mathbf{\Lambda}$ may be dense, so the block-diagonal factorization used above is not automatically available. The useful case is when most of the correlation remains local to partitions. We then separate the information matrix into

$$\mathbf{G}^{-1} = \mathbf{G}_{\text{on}}^{-1} + \mathbf{G}_{\text{off}}^{-1},$$

where $\mathbf{G}_{\text{on}}^{-1}$ collects the within-partition block-diagonal terms and $\mathbf{G}_{\text{off}}^{-1}$ collects the remaining cross-partition terms. When $\mathbf{G}_{\text{off}}^{-1}$ is sparse and low rank with respect to the partition structure, Woodbury's identity can be leveraged to preserve the transformed-OLS formulation. This occurs, for example, when there are many partitions but each correlated cluster overlaps only a few of them. In the limiting case where each correlated cluster stays within a single partition, $\mathbf{G}_{\text{off}}^{-1} = \mathbf{0}$ and the independent-case block factorization is recovered exactly. More explicitly,

$$\mathbf{G}^{-1} = (\mathbf{X}^{\top} \mathbf{X} + \mathbf{\Lambda}) + \mathbf{X}^{\top} (\mathbf{V}^{-1} - \mathbf{I}) \mathbf{X} = \mathbf{G}_{\text{on}}^{-1} + \mathbf{G}_{\text{off}}^{-1}.$$

The inverse of the remaining off-block-diagonal, cross-partition term is written as a reduced spectral factorization, $\mathbf{G}_{\text{off}}^{-1} = \mathbf{E} \mathbf{J} \mathbf{E}^{\top}$, where $\mathbf{E}$ absorbs the absolute eigenvalue magnitudes and $\mathbf{J}$ records their signs over the s retained eigen-directions. Woodbury's identity gives

$$\begin{aligned} \mathbf{G} &= \mathbf{G}_{\text{on}} - \mathbf{G}_{\text{on}} \mathbf{E} (\mathbf{J}^{-1} + \mathbf{E}^{\top} \mathbf{G}_{\text{on}} \mathbf{E})^{-1} \mathbf{E}^{\top} \mathbf{G}_{\text{on}} \\ &= \mathbf{G}_{\text{on}}^{1/2} \mathbf{F} \mathbf{G}_{\text{on}}^{1/2}, \end{aligned}$$

requiring only an $s \times s$ inverse for the computation. Supplement B shows how $\mathbf{F}^{1/2}$ and $\mathbf{F}^{-1/2}$ are obtained from a QR decomposition of $\mathbf{G}_{\text{on}}^{1/2} \mathbf{E}$. The transformed-OLS step is then preserved through a factor $\mathbf{L} = \mathbf{G}_{\text{on}}^{1/2} \mathbf{F}^{1/2}$ satisfying $\mathbf{L} \mathbf{L}^{\top} = \mathbf{G}$ in place of $\mathbf{G}^{1/2}$. Thus, many computationally efficient block-diagonal operations of the independent case remain available, although the construction of $\mathbf{F}$ reduces the memory savings relative to the independent case.

2.6 Bayesian Inference

A Bayesian interpretation of monomial splines is straightforward by interpreting smoothing constraints as an operation applied to the otherwise unconstrained posterior of model parameters. For illustration with the canonical normal-normal conjugate scenario with known σ^2 and the following prior and conditional distributions respectively:

$$\boldsymbol{\beta} | \sigma^2 \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{\Lambda}^{-1}), \quad (12)$$

$$\mathbf{y} | \mathbf{X}, \boldsymbol{\beta}, \sigma^2 \sim \mathcal{N}(\mathbf{X} \boldsymbol{\beta}, \sigma^2 \mathbf{I}). \quad (13)$$

The unconstrained posterior distribution follows the usual normal-normal conjugacy identity:

$$\boldsymbol{\beta}|\mathbf{y}, \mathbf{X}, \sigma^2 \sim \mathcal{N}(\hat{\boldsymbol{\beta}}, \sigma^2 \mathbf{G}). \quad (14)$$

The posterior distribution under constraint can be expressed as follows for $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$\boldsymbol{\beta}|\mathbf{y}, \mathbf{X}, \sigma^2, \mathbb{I}_{\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{0}} \equiv \tilde{\boldsymbol{\beta}} + \sigma \mathbf{U} \mathbf{G}^{\frac{1}{2}} \mathbf{z} \equiv \mathbf{U}(\hat{\boldsymbol{\beta}} + \sigma \mathbf{G}^{1/2} \mathbf{z}). \quad (15)$$

Thus draws of $\tilde{\boldsymbol{\beta}}$ (which follows a P -dimensional degenerate distribution) can be obtained by simply left-multiplying draws of $\hat{\boldsymbol{\beta}}$ (which follows a non-degenerate P -dimensional distribution) by $\mathbf{U}$. If construction of $\mathbf{U}$ is to be avoided, the same OLS computational technique in 2.5.1 can be applied by using a draw of $\boldsymbol{\beta}$ from its unconstrained posterior in place of $\hat{\boldsymbol{\beta}}$. Wood (2006) discusses how even with tuning, $\frac{1}{\sigma^2} \mathbf{\Lambda}$ can be treated as a prior precision matrix for $\boldsymbol{\beta}$.

Several choices are made for implementation in `lgspline`, but are not essential to the procedure. A Laplace approximation is implemented for non-Gaussian response, replacing $\mathbf{G}$ with $\tilde{\mathbf{G}}(\tilde{\boldsymbol{\beta}})$ as in 2.5.2. When inequality constraints are active, elliptical slice sampling (Murray, Adams, and MacKay, 2010) can be used to retain only draws in the feasible region respecting inequality constraints. When correlation parameters are modeled, their uncertainty may be propagated by drawing from a normal approximation centered at the REML estimate, while σ^2 may be drawn from an inverse-gamma distribution derived as the marginal posterior given the observed data and smoothing constraints; in `lgspline` the default prior distribution for σ^2 is `InvGamma(0,0)`. Once σ^2 and correlation parameters are drawn from their posteriors, $\mathbf{G}$ can be re-constructed conditional upon them and in turn, used to generate posterior draws of coefficients using the approach above. This is demonstrated in Section 3.2.

2.7 Practical Considerations

The monomial framework is general, but any implementation requires choices regarding partitioning, scaling, and handling many non-spline effects simultaneously. We briefly

describe the defaults used in `lgspline`, however, these are not the only valid choices, and the implementation allows all to be customized.

Knot locations are determined by k -means clustering on a standardized predictor matrix, returning centers on the raw scale, with midpoints between neighboring centers serving as partition boundaries. Custom knots, boundaries, and partitioning schemes can be alternatively supplied by the user.

Polynomial basis expansions are scaled by $q_{0.69} - q_{0.31}$, where q_ζ is the ζ th quantile of the expansion, approximately one standard deviation under normality. This robust scaling avoids sensitivity to outliers while keeping higher-order terms on a comparable scale. The penalty matrices and constraints are all scaled by the same values to be consistent. Back-transforming for standard errors and coefficient estimates is computationally negligible, since it only involves rescaling by the same value.

For models with many non-spline effects and relatively few spline effects, a modified backfitting approach (Hastie and Tibshirani, 1986) alternates between spline-effect estimation using the procedures above and non-spline estimation using standard maximum likelihood. While first-derivative constraints in theory constrain all non-spline effects across partitions to be equivalent (coefficients *are* first derivatives of linear expansions), in practice for q^* non-spline effects there are at least Kq^* redundant coefficients that need to be estimated and is thus computationally inefficient. The modified backfitting leaves point estimates and standard error estimates unchanged and ensures that all non-spline effects remain constant across partitions more efficiently than constrained optimization directly.

2.8 Software and hardware.

All analyses were implemented in R version 4.5.1 (R Core Team, 2025). The reported results use `lgspline` version 1.1.0; the `mgcv` comparison fits use version 1.9-3. Session information was printed and saved with the output files for each result-generating script. Timing analyses were run under Windows 11 on an ASUS Vivobook laptop with a 13th Gen Intel Core i5-1334U processor (10 physical cores, 12 logical processors) and 8 GB RAM. Parallel examples used a 12-worker cluster.

3 Results

We report three complementary evaluations. A simulation study across three data-generating processes compares predictive accuracy and runtime against established spline implementations. A pharmacokinetic application emphasizes interpretability and shows how the monomial framework supports shape constraints, structured correlation, and derived parameter inference in a single fit. A parallelism benchmark examines runtime scaling with the number of partitions.

3.1 Simulation Study

Data-generating processes. Three data-generating processes (DGPs) of increasing dimension were chosen to exercise different aspects of multivariate spline fitting under less-than-ideal conditions. The univariate target is a standard nonparametric regression benchmark (Donoho and Johnstone, 1994) with spatially varying frequency and early oscillatory behavior that is difficult to model. The bivariate $q = 2$ and four-dimensional $q = 4$ scenarios use predictors with skewed margins, unequal ranges, mild mixture contamination, and surfaces with smooth oscillations, localized Gaussian features, three-way interactions, and light nonsmooth terms. Predictor ranges differ by more than an order of magnitude, which would otherwise favor methods that internally rescale, and the mixture contamination produces small high-density regions that make basis dimension selection non-trivial. These choices avoid the more standard setting of independent uniform predictors on a common scale, where most spline methods perform comparably and meaningful differences are hard to see. Exact formulas and surface visualizations are provided in Supplement E.

Spline Methods. Four methods are compared for each DGP. The first is the monomial formulation as implemented in `lgspline`, fit with up-to-quartic univariate terms, smoothness constraints, and closed-form curvature penalty. The second is a cubic regression spline with cubic smoothing spline penalty (`bs = "cr"`) via `mgcv`. For $q \geq 2$, the cubic regression spline includes main effects and interaction terms through `te()`, with three-way interactions added when $q = 4$. The third method is a thin-plate regression spline (TPRS; `bs =`

"tp") via `mgcv`, fit as a joint smooth of all predictors, with the knot subsample used in the thin-plate construction capped at 2,000 for large N . The fourth method is a truncated quartic polynomial spline with a ridge penalty on the truncated terms, constructed manually. All four methods are fit after centering and scaling predictors within each replicate. For `lgspline` and the truncated polynomial fit, the interaction convention includes linear two-way interactions, linear-quadratic interactions, and linear three-way interactions for $q = 4$.

K^* is used to roughly quantify the increase in number of partitions per predictor, although it does not have exactly the same meaning for every method. For `lgspline`, K^* gives the number of partitions per predictor through $K = K^*q$. For the cubic regression spline, K^* is the marginal basis dimension for main effects, with lower-dimensional interaction bases determined from it. For TPRS, the basis dimension is K^* for $q = 1$, $k = 27$ for $q = 2$, and $k = 100$ for $q = 4$. For the truncated polynomial, K^* is the number of interior knots per predictor. We use $K^* = 16$ for $q = 1$, $K^* = 8$ for $q = 2$, and $K^* = 4$ for $q = 4$. These choices are meant to keep the fits computationally reasonable for reproducibility while still giving the lower-dimensional methods enough resolution to capture local structure.

Sample sizes are $N \in \{500, 5,000, 50,000\}$ for each q , with Gaussian noise at $\sigma \in \{0.5, 2.0\}$. The tuned regime selects smoothness by REML for `mgcv`, the default constrained leave-one-out criterion for `lgspline`, and `mgcv::magic` for the truncated polynomial. In the fixed-penalty regime, smoothing parameters are fixed at 10^{-4} for `mgcv` and the truncated polynomial, while `lgspline` uses its defaults $\lambda_w = 2 \times 10^{-7}$ and $\lambda_r = 0.5$. Each scenario has $M = 24$ replications with deterministic per-task seeds. The primary metric is $\text{RMSE} = \|\tilde{f} - f\|_2 / \sqrt{N}$, comparing the estimated function $\tilde{f}$ with the true data-generating function f at the observed predictor values. Fitting time under tuning is considered secondary. All methods use the same centered and scaled predictors prior to fitting.

Performance across DGPs. Table 1 reports mean RMSE under automatic smoothness selection; fixed-penalty RMSE appears in Supplement F. Tuning criteria and optimization techniques differ, the meaning of K^* differs by method, and `lgspline` and the truncated

polynomial include quartic terms while Cubic RS does not, so the methods are not directly comparable. These simulations should also not be read as a universal ranking of smoothers; different data-generating functions, tuning criteria, and software implementations would change the comparison. The purpose is to show the proposed method remains competitive across several nontrivial settings while retaining a directly interpretable local polynomial representation.

At $q = 1$, `lgspline` has the lowest RMSE in every tuned cell. At $N = 50,000$ and $\sigma = 0.5$, RMSE is 0.463 for `lgspline`, compared with 0.692 for Cubic RS, 0.716 for TPRS, and 0.688 for the truncated polynomial; at $\sigma = 2.0$, the corresponding values are 0.451, 0.698, 0.717, and 0.694. The improvement is large, and the additional tuned runtime at smaller N is less than half a second compared with Cubic RS. The first high-frequency oscillations near $t = 0$ remain difficult to capture (Figure 1), but this difficulty is shared across the fitted methods at this basis dimension.

At $q = 2$, `lgspline` is best in four of six tuned cells, including all three cells at $\sigma = 0.5$ and the largest sample size at $\sigma = 2.0$. At $N = 50,000$ and $\sigma = 0.5$, RMSE is 0.104 for `lgspline`, 0.319 for Cubic RS, and 0.413 for the truncated polynomial, separations of roughly a factor of three and four respectively. TPRS is best for the two smaller, noisy samples. In those two cells, the RMSE advantage of TPRS is modest (0.069 at $N = 500$ and 0.029 at $N = 5,000$ with $\sigma = 2$), and `lgspline` catches up at $N = 50,000$.

At $q = 4$, Cubic RS is most accurate in four of six tuned cells, TPRS is best at $N = 500$, $\sigma = 2.0$, and `lgspline` is best at the largest low-noise sample. At $N = 50,000$ and $\sigma = 0.5$, RMSE is 0.108 for `lgspline`, 0.122 for Cubic RS, 0.174 for TPRS, and 0.182 for the truncated polynomial. At $\sigma = 2.0$, Cubic RS reaches 0.165, compared with 0.186 for `lgspline`, 0.193 for TPRS, and 0.193 for the truncated polynomial.

The $q = 4$ results show Cubic RS can be slightly more accurate, but `lgspline` scales more efficiently with N . For example, at $N = 5,000$ and $\sigma = 0.5$, its RMSE is only 0.0066 larger than Cubic RS while fitting in 6.6 seconds rather than 266 seconds; at $N = 50,000$ and $\sigma = 2.0$, its RMSE is just over 0.02 units greater while fitting in 8.1 seconds rather than 1,631 seconds. In the largest low-noise $q = 4$ cell, the fixed-penalty `lgspline` fit in

Supplement F is essentially identical to the tuned fit (RMSE 0.107 versus 0.108), a useful reminder that smoothing selection is not guaranteed to improve every finite-sample average.

Timing. Figure 2 and Tables 2–3 give wall-clock time. At small N and low q , Cubic RS and TPRS are usually slightly faster than **lgspline**. In the $q = 1$ and $q = 2$ cells where **lgspline** is slower than another spline-software fit, the gap is no more than about 1.2 seconds. Under automatic tuning, the truncated polynomial is fastest in every cell. This should be interpreted alongside its simpler dense least-squares implementation and tuning setup, rather than as a general property of truncated power bases. Among the spline-software fits, low-rank TPRS is fast at $q = 2$ and $q = 4$, but **lgspline** is faster than both **mgcv** smoothers in the largest $q = 4$ samples. For $N = 50,000$ and $\sigma = 0.5$, **lgspline** takes 7.0 seconds, compared with 48.8 seconds for TPRS and 2,565 seconds for Cubic RS, a roughly 7-fold and 370-fold reduction. With penalties fixed, the block-diagonal advantage is even clearer; at $q = 4$, $N = 50,000$, $\sigma = 0.5$, mean times are 2.5 seconds for **lgspline**, 26.2 seconds for TPRS, and 173 seconds for Cubic RS. The same pattern holds at $\sigma = 2$, where Cubic RS crosses 200 seconds while **lgspline** stays under three seconds.

Table 1: Mean RMSE against f_{true} under automatic smoothness selection over 24 replications (SD in parentheses). Lowest RMSE in each column is **bold**. Fixed-penalty results appear in Supplement F.

σ	Method	$q = 1$			$q = 2$			$q = 4$		
		$N=500$	$N=5k$	$N=50k$	$N=500$	$N=5k$	$N=50k$	$N=500$	$N=5k$	$N=50k$
0.5	lgspline	0.477 (0.043)	0.468 (0.032)	0.463 (0.032)	0.229 (0.015)	0.121 (0.006)	0.104 (0.004)	0.349 (0.087)	0.158 (0.004)	0.108 (0.002)
	Cubic RS	0.714 (0.068)	0.703 (0.027)	0.692 (0.013)	0.334 (0.011)	0.319 (0.004)	0.319 (0.002)	0.226 (0.010)	0.152 (0.008)	0.122 (0.001)
	TPRS	0.714 (0.045)	0.724 (0.010)	0.716 (0.003)	0.240 (0.015)	0.212 (0.006)	0.211 (0.003)	0.237 (0.011)	0.183 (0.004)	0.174 (0.003)
	Trunc. poly	0.673 (0.064)	0.696 (0.021)	0.688 (0.011)	0.425 (0.013)	0.413 (0.005)	0.413 (0.002)	0.228 (0.011)	0.189 (0.003)	0.182 (0.001)
2.0	lgspline	0.686 (0.051)	0.487 (0.034)	0.451 (0.030)	0.509 (0.053)	0.284 (0.061)	0.135 (0.004)	0.587 (0.055)	0.354 (0.017)	0.186 (0.005)
	Cubic RS	0.809 (0.058)	0.710 (0.039)	0.698 (0.014)	0.478 (0.047)	0.346 (0.007)	0.321 (0.002)	0.487 (0.067)	0.254 (0.015)	0.165 (0.010)
	TPRS	0.803 (0.039)	0.730 (0.012)	0.717 (0.003)	0.440 (0.051)	0.256 (0.008)	0.215 (0.004)	0.462 (0.056)	0.280 (0.011)	0.193 (0.004)
	Trunc. poly	0.774 (0.055)	0.700 (0.037)	0.694 (0.013)	0.521 (0.038)	0.430 (0.008)	0.415 (0.001)	0.571 (0.070)	0.264 (0.017)	0.193 (0.002)

3.2 Application to Pharmacokinetics

We illustrate several features of **lgspline** on the Theophylline pharmacokinetic dataset in base R (Boeckmann, Sheiner, and Beal, 1994; R Core Team, 2025), with 132 observations

Table 2: Mean wall-clock time in seconds under automatic smoothness selection over 24 replications (SD in parentheses). Fastest method in each column is **bold**.

σ	Method	$q = 1$			$q = 2$			$q = 4$		
		500	5k	50k	500	5k	50k	500	5k	50k
0.5	lgspline	0.67 (0.22)	0.71 (0.09)	1.88 (0.30)	1.27 (0.46)	1.31 (0.21)	4.11 (0.78)	10.03 (3.21)	6.56 (1.59)	6.99 (1.94)
	Cubic RS	0.23 (0.19)	0.47 (0.23)	5.64 (1.80)	0.44 (0.23)	1.37 (0.32)	22.40 (10.31)	23.16 (8.27)	266.45 (142.77)	2564.82 (922.35)
	TPRS	0.38 (0.20)	1.74 (0.36)	16.54 (4.43)	0.30 (0.17)	2.81 (0.43)	20.58 (3.50)	1.08 (0.17)	6.75 (1.22)	48.77 (8.76)
	Trunc. poly	0.09 (0.12)	0.04 (0.06)	0.14 (0.04)	0.07 (0.09)	0.03 (0.04)	0.18 (0.08)	0.03 (0.01)	0.04 (0.02)	0.20 (0.05)
2.0	lgspline	0.52 (0.16)	0.71 (0.14)	1.87 (0.25)	1.41 (0.55)	1.35 (0.35)	4.14 (0.76)	11.02 (4.77)	7.70 (2.56)	8.06 (2.21)
	Cubic RS	0.16 (0.13)	0.44 (0.11)	5.91 (2.17)	0.46 (0.26)	1.49 (0.46)	23.32 (7.22)	22.49 (8.55)	149.19 (58.28)	1631.27 (483.28)
	TPRS	0.38 (0.24)	1.64 (0.29)	15.13 (4.03)	0.24 (0.12)	2.85 (0.44)	19.14 (2.83)	0.99 (0.20)	8.55 (2.33)	66.19 (10.44)
	Trunc. poly	0.06 (0.09)	0.04 (0.06)	0.13 (0.05)	0.03 (0.03)	0.03 (0.05)	0.16 (0.08)	0.02 (0.01)	0.04 (0.01)	0.23 (0.05)

Table 3: Mean wall-clock time in seconds under the fixed-penalty regime over 24 replications (SD in parentheses). Fastest method in each column is **bold**.

σ	Method	$q = 1$			$q = 2$			$q = 4$		
		500	5k	50k	500	5k	50k	500	5k	50k
0.5	lgspline	0.08 (0.03)	0.18 (0.02)	1.13 (0.13)	0.16 (0.04)	0.34 (0.06)	2.89 (0.37)	0.50 (0.07)	0.59 (0.09)	2.46 (0.45)
	Cubic RS	0.15 (0.12)	0.35 (0.15)	3.26 (1.20)	0.21 (0.10)	0.64 (0.14)	8.75 (2.72)	3.77 (0.65)	23.78 (5.96)	173.43 (28.52)
	TPRS	0.16 (0.11)	1.10 (0.20)	5.92 (1.58)	0.19 (0.10)	2.66 (0.42)	17.56 (2.23)	0.49 (0.09)	4.65 (0.64)	26.19 (4.81)
	Trunc. poly	0.00 (0.00)	0.00 (0.01)	0.04 (0.02)	0.00 (0.00)	0.00 (0.01)	0.06 (0.03)	0.00 (0.01)	0.01 (0.01)	0.06 (0.02)
2.0	lgspline	0.08 (0.02)	0.19 (0.03)	1.16 (0.12)	0.13 (0.04)	0.35 (0.04)	2.90 (0.45)	0.50 (0.09)	0.64 (0.13)	2.66 (0.40)
	Cubic RS	0.17 (0.12)	0.42 (0.17)	3.92 (1.30)	0.26 (0.15)	0.75 (0.15)	9.89 (2.69)	4.46 (0.65)	34.10 (7.02)	227.19 (27.57)
	TPRS	0.20 (0.19)	1.19 (0.26)	6.41 (1.76)	0.26 (0.13)	2.71 (0.49)	18.48 (2.68)	0.57 (0.10)	5.50 (0.75)	32.53 (5.36)
	Trunc. poly	0.00 (0.01)	0.00 (0.01)	0.04 (0.02)	0.00 (0.00)	0.00 (0.01)	0.05 (0.02)	0.00 (0.00)	0.01 (0.01)	0.06 (0.01)

from 12 subjects measured at 11 irregularly spaced time points over approximately 24 hours post-dose. The aim is to demonstrate biologically motivated model construction, structured covariance, simultaneous shape constraints, and direct extraction of non-compartmental pharmacokinetic summaries (Gabrielsson and Weiner, 2016) with fully propagated uncertainty in a single fit.

Serum concentration was modeled as a quadratic spline in time with a linear effect of weight-normalized dose and a dose-by-time quadratic interaction, using $K = 10$ interior knots placed between consecutive observed time points, yielding 11 partitions, one for each unique timepoint sampled per subject. Model structure was chosen to reflect biological plausibility and to illustrate features of **lgspline** rather than selected by AIC or cross-

validation. Specifically, within each partition the model takes the form

$$\tilde{f}(\text{Time}, \text{Dose}) = \beta_0 + \beta_1 \text{Time} + \beta_2 \text{Time}^2 + \beta_3 \cdot \text{Dose} + \beta_4 \cdot \text{Dose} \cdot \text{Time}^2,$$

with Time in post-dose hours and Dose as weight-normalized Theophylline dosage in mg/kg. The quadratic-linear interaction lets dose modulate curvature through one interpretable parameter. Excluding the linear Time $\times$ Dose interaction makes the initial absorption rate dose-independent while allowing the initial acceleration to vary with dose. In this parameterization, β_3 has units kg/L and β_4 has units kg/(L $\cdot$ h²), so both dose terms contribute on the concentration scale after multiplying by dose in mg/kg. Continuity and first-derivative smoothness were imposed; second-derivative constraints were excluded, as is typical for a quadratic spline.

Within-subject correlation was modeled via a customized tridiagonal working correlation matrix with a common estimated parameter ρ , such that only neighboring observations with respect to time are correlated within subject (the first timepoint is correlated with the second timepoint for each subject, but not the third timepoint, nor between subjects). This structure was chosen to capture the dominant serial dependence between temporally adjacent measurements without using a dense exchangeable working correlation. The correlation parameter was initialized at $\tanh^{-1}(0.1)$, corresponding to a weak positive adjacent-time correlation, which was found to give stable REML optimization for this structured covariance model. It was then estimated by maximizing the REML profile likelihood using the structured covariance extension described in Section 2.5.2. Three simultaneous inequality constraints were imposed for biological plausibility: a non-positive first derivative with respect to time for Time > 5 hours, enforcing monotone elimination; a non-negative second derivative for Time > 5 hours, enforcing convexity during terminal elimination; and non-negative fitted concentration for Time ≥ 0 , since negative concentration is impossible. The time threshold of 5 hours was chosen to exclude the absorption and early distribution phase.

A notable feature of the fit is that the quadratic dose-time interaction is constant across partitions, a consequence of the smoothness constraints and this dataset. This reduces the number of partition-specific relationships and makes the printed equations easier to read.

The leftmost partition, covering the absorption phase, yields the closed-form quadratic polynomial

$$\tilde{f}(\text{Time}, \text{Dose}) = -0.8707 + 6.7865 \text{ Time} + 6.1856 \text{ Time}^2 + 0.2809 \cdot \text{Dose} + 0.0001 \cdot \text{Dose} \cdot \text{Time}^2.$$

The first derivative at $\text{Time} = 0$ is $\tilde{f}'(0, \text{Dose}) = 6.7865 \text{ mg}/(\text{L} \cdot \text{h})$, giving the estimated initial absorption rate directly from the monomial coefficients. The second derivative at $\text{Time} = 0$ is $\tilde{f}''(0, \text{Dose}) = 2 \times 6.1856 + 2 \times 0.0001 \times \text{Dose} = 12.3712 + 0.0002 \cdot \text{Dose} \text{ mg}/(\text{L} \cdot \text{h}^2)$, representing the initial acceleration of absorption, which increases slightly with dosage. Both quantities are available in closed form in every partition and can be evaluated without post-hoc basis conversion. The printed equations in Figure 3 give the fitted local response model directly.

For posterior inference, let $\tilde{f}^{(m)}$ be the concentration-time curve at dose d implied by the m th posterior draw $\tilde{\beta}^{(m)}$. We generated $M = 1,000$ draws as in Section 2.6, satisfying smoothness constraints without imposing inequality constraints. The correlation parameter ρ was drawn from a normal approximation on the $\tanh^{-1}$ scale and back-transformed. The variance parameter σ^2 was drawn from its `lgspline` default marginal posterior, $\sigma^2 | \mathbf{y}, \mathbf{X} \sim \text{InvGamma}\{(N - \text{tr}(\mathbf{H}))/2, (N - \text{tr}(\mathbf{H}))\tilde{\sigma}^2/2\}$. The six pharmacokinetic summaries in Table 4 follow standard non-compartmental definitions (Gabrielsson and Weiner, 2016). AUC at 24 hours was computed by Gauss–Legendre quadrature. Draws with $\text{AUC}_{0-24} \leq 0$ or $\text{AUC}_{0-24}/\text{AUC}_{\infty}$ ratio ≥ 1 were excluded as biologically implausible; only 0.9% of draws were excluded.

Figure 3 summarizes the analysis. The fitted curve rises sharply through absorption and transitions to monotone convex decay after $\text{Time} = 5$ hours. The uncertainty band is tight near dense observed time points and widens late in follow-up, where the terminal curve is carried by fewer measurements. The posterior for $t_{\max}$ is visibly bimodal, with a primary early mode and a smaller later mode, matching the two-peaked appearance of the raw data during absorption and peak concentration. Heavy tails on AUC_{0-24} , $t_{1/2}$, and λ_z reflect the sensitivity of terminal-phase extrapolation to the short observation window.

The point estimate of the correlation parameter is 0.235, and a 95% confidence interval based on the inverse Hessian of the BFGS optimization implemented in the `optim` function

Table 4: Posterior-draw definitions of the pharmacokinetic summaries computed from the fitted concentration-time curve $\tilde{f}^{(m)}(t; d)$ at dose d .

Name	Symbol	Description	Formula
Time to peak concentration	$t_{\max}^{(m)}$	Time at which the fitted concentration attains its maximum on $[0, 24]$	$\arg \max_{0 \leq t \leq 24} \tilde{f}^{(m)}(t; d)$
Peak concentration	$C_{\max}^{(m)}$	Maximum fitted concentration	$\tilde{f}^{(m)}(t_{\max}^{(m)}; d)$
Half-maximum time	$t_{1/2}^{(m)}$	First post-peak time at which the fitted concentration falls to one half of $C_{\max}^{(m)}$	$\inf\{t > t_{\max}^{(m)} : \tilde{f}^{(m)}(t; d) = C_{\max}^{(m)}/2\}$
Terminal elimination rate constant	$\lambda_z^{(m)}$	Log-linear elimination slope implied by the peak and half-maximum times	$\frac{\log(0.5)}{t_{\max}^{(m)} - t_{1/2}^{(m)}}$
Exposure through 24 hours	$\text{AUC}_{0-24}^{(m)}$	Area under the fitted concentration-time curve on the observed window	$\int_0^{24} \tilde{f}^{(m)}(t; d) dt$
Exposure fraction observed by 24 hours	$\text{AUC}_{\text{ratio}}^{(m)}$	Fraction of extrapolated total exposure captured by 24 hours	$\frac{\text{AUC}_{0-24}^{(m)}}{\text{AUC}_{0-24}^{(m)} + \tilde{f}^{(m)}(24; d)/\lambda_z^{(m)}}$

of base R (R Core Team, 2025) for REML gives bounds of 0.232 to 0.239. The fitted adjacent-time correlation is therefore positive in this example and suggests a modest but detectable intrasubject correlation explaining variability in the data, which is biologically plausible.

Table 5 reports posterior quantiles and 95% credible intervals at normalized Dose = 5 mg/kg. The expected time to peak concentration has a posterior median of 1.41 hours with a 95% CI of 1.22 to 2.86, consistent with rapid absorption. The expected maximum concentration has a median of 8.46 mg/L with a 95% CI of 7.43 to 9.52. Elimination half-life has a median of 11.09 hours with a 95% CI of 8.73 to 14.28, and the terminal elimination rate constant has a median of 0.073 h⁻¹ with a 95% CI of 0.055 to 0.100, indicating a terminal decline that is appreciable but not abrupt over the 24-hour follow-up. Total drug exposure has a median AUC_{0-24} of 100.8 mg·h/L with a 95% CI of 32.9 to 171.2. The wide interval mostly reflects uncertainty in unobserved or sparsely observed portions of the curve rather than uncertainty near the observed concentration peak. The $\text{AUC}_{0-24}/\text{AUC}_{\infty}$ ratio has a median of 0.81 with a 95% CI of 0.62 to 0.92, suggesting that for the majority of

Table 5: Posterior pharmacokinetic summary distributions from `lgspline` ($M = 1000$ draws total, 991 draws retained after excluding biologically implausible draws, normalized Dose = 5 mg/kg).

Parameter	Min	2.5%	25%	Median	75%	97.5%	Max
$t_{\max}$ (hours)	0.78	1.22	1.34	1.41	1.84	2.86	3.90
$C_{\max}$ (mg/L)	6.97	7.43	8.12	8.46	8.80	9.52	10.32
$t_{1/2}$ (hours)	7.11	8.73	10.23	11.09	11.99	14.28	24.64
λ_z (h^{-1})	0.030	0.055	0.066	0.073	0.081	0.100	0.124
AUC ₀₋₂₄ (mg·h/L)	0.87	32.87	76.34	100.83	126.29	171.21	224.28
AUC-24/AUC- ∞	0.04	0.62	0.76	0.81	0.85	0.92	1.00

subjects a meaningful fraction of total exposure occurs beyond 24 hours.

The combination of user-specified structured within-subject correlation, simultaneous arbitrary shape constraints, and a biologically motivated model built directly in monomial form is not, to our knowledge, jointly available in a single fit in existing R spline software. These results demonstrate that the monomial framework recovers clinically interpretable PK summaries with fully propagated posterior uncertainty in a single integrated fit, requiring no post-hoc algebraic basis conversion nor assumption of an underlying compartmental model.

3.3 Evaluating Parallelism

To assess the computational behavior of the block-diagonal structure under parallelism, `lgspline` was fit to the Maunga Whau volcano dataset (R Core Team, 2025), an 87×61 grid yielding $N = 5,307$ observations with $q = 2$ spatial dimensions as predictors and elevation as the response. For RMSE calculation, a deterministic 10% holdout sample was used and the reported times are for fitting on the remaining observations. Five partition counts were considered: $K + 1 \in \{100, 250, 500, 750, 1000\}$. Each configuration was fit five times with and without a 12-core parallel cluster, using fixed penalty values, default polynomial expansions (including quartic main effects for this two-predictor fit), no derivative

constraints, and quadratic interactions. Models were not tuned, because different tuning runs have different numbers of iterations and introduce variance not attributable to parallelism. This isolates the default fit-time parallelism available after penalty parameters have been fixed. Parallel processing was applied to neighbor determination for each partition, smoothness constraint setup, and the eigenvalue decompositions used to compute the square root and inverse square root of $\mathbf{G}$.

Figure 4 shows the measured speedup, defined as sequential time divided by parallel time. At $K + 1 = 100$, parallel overhead dominates and sequential fitting is faster (speedup $0.33\times$). Parallelism is approximately break-even at $K + 1 = 250$ (speedup $1.02\times$), and then remains modest: $1.25\times$ at $K + 1 = 500$ and $1.22\times$ at $K + 1 = 1,000$. These gains are smaller than hoped, and suggest that the serial QR decompositions involving $\mathbf{A}$ and $\mathbf{G}^{1/2}\mathbf{A}$ become important bottlenecks. Larger gains in speed will likely require parallelizing more of the remaining linear algebra or using hardware with more substantial parallel compute.

Figure 5 shows an illustrative tuned fit with $K + 1 = 250$ for context.

Table 6: Wall-clock time in seconds for `lgspline` on the volcano dataset (87×61 grid, $N = 5,307$, $q = 2$; 10% holdout for RMSE, fitting on the remaining 4,777 observations), fixed penalty values, mean $\pm$ SD over 5 replications.

$K+1$	Sequential		Parallel (12 cores)		Speedup		RMSE	
100	1.5	(0.1)	5.5	(3.0)	0.33	(0.15)	1.19	(0.02)
250	21.7	(0.1)	21.5	(2.6)	1.02	(0.11)	0.94	(0.08)
500	224.4	(0.4)	179.0	(4.5)	1.25	(0.03)	0.94	(0.06)
750	933.0	(2.5)	746.1	(6.3)	1.25	(0.01)	0.86	(0.03)
1000	2229.4	(5.5)	1824.0	(30.1)	1.22	(0.02)	0.87	(0.01)

4 Discussion

The main contribution of this article is a computational strategy for fitting penalized polynomial splines in monomial form using constrained optimization. The method is designed

for settings where a local polynomial representation is useful for interpretation. The transformed OLS formulation exploits the block-diagonal partition structure for both memory savings and faster computation, while also creating a natural path to parallel execution. This computational convenience is retained under GLM weights, inequality constraints including shape constraints, and certain settings with correlated response.

Relative to truncated polynomial splines, the proposed method keeps the representation local. Each partition depends on only p coefficients rather than potentially referencing the full P -dimensional coefficient vector, and the full dense linear system never needs to be assembled. The truncated power basis is simple to construct after predictor standardization, but its fitted equations are not local. Relative to B-spline methods, the method trades some numerical conditioning for direct interpretability.

The simulation study of Section 3.1 demonstrates its competitive performance and runtime versus other spline constructions. The implementation of `lgspline` is strongest at $q = 1$ and in most $q = 2$ cells, while Cubic RS and TPRS remain strong at $q = 4$, especially in smaller, noisier samples. Under tuned and fixed penalties, `lgspline` is often faster at larger N relative to tensor-product cubic regression splines and thin-plate splines. The truncated polynomial basis was usually less accurate but fastest in these runs, a result that should be read in light of its simple dense implementation and tuning setup. These simulations are limited, and performance would change on other datasets. They nevertheless show the main practical point; the monomial formulation can give competitive fits, can be much faster as sample size grows, and does not introduce an obvious computational cost for the improved interpretability.

The pharmacokinetic application shows the practical value of the method for carefully constructed parametric models of predictor-response relationships, and how the monomial formulation enables certain combinations of modeling features that are difficult to assemble using other spline frameworks or software.

The volcano example shows that although many of the model fitting operations are parallelizable in principle, the current implementation gives only modest speedups using the hardware here. We expect larger gains to be realized with hardware that can better

leverage parallelism, but we did not have the resources to show this here.

There are clear limitations to the proposed method. Smoothness and shape constraints are satisfied only to numerical tolerance, which can be checked by computing $\mathbf{A}^\top \tilde{\boldsymbol{\beta}}$ (or $\mathbf{A}^\top \tilde{\boldsymbol{\beta}} - \mathbf{c}$ for non-zero constraints). Over-constrained models can force the spline toward a single polynomial shared across all partitions. The default heuristic sums first- and second-derivative constraints to avoid this behavior in many cases, but the resulting fit may no longer be exactly differentiable between partitions. Inequality constraints are enforced at observed data points only, unlike the global pointwise enforcement of B-spline shape-constrained implementations such as `scam` (Pya and Wood, 2015); sparse regions may need additional user-specified constraint points. With multiple predictors, the number of partition neighbors can grow quickly, increasing the constraint system and reducing some computational advantages. An additive decomposition in which smoothness constraints operate independently per predictor would help, but is not currently implemented. Penalization is also effectively necessary when the partition count is large relative to N ; otherwise the unconstrained per-partition fits used by the transformed-OLS procedure can become degenerate. Finally, piecewise polynomials require more care to interpret than a single global linear term. This limitation applies to polynomial splines generally, although the fitted function within each partition is still expressible without knot locations or basis conversion in arguably the simplest form a polynomial can take.

We argue in favor of using the proposed monomial spline in practice, but it has intrinsic value as an educational device alone. It is a rich application of constrained optimization to statistics, and many students may appreciate seeing a polynomial in a familiar form before moving on to more complicated basis constructions, which is in agreement with Ezhov, Neitzel, and Petrovic (2018). Even a reader who prefers B-splines for their numerical properties may find the monomial formulation useful as an educational complement.

Potential future work includes quantile regression and location-scale extensions in the spirit of `qgam` (Fasiolo, Wood, Zaffran, Nedellec, and Goude, 2021) and `gamlss` (Stasinopoulos and Rigby, 2007), and more aggressive high-performance computing or GPU implementations to take advantage of parallelization. More broadly, the same transformed

constrained-OLS framework may be useful beyond spline fitting whenever a problem has block-diagonal design structure and linear constraints. Software implementing all procedures described here is available in `lgspline` for R on CRAN.

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Supplementary Materials

Code: R scripts reproducing all simulations and applications, available at <https://github.com/matthewlouisdavisBioStat/lgspline>. Reported results correspond to `lgspline`

version 1.1.0. The scripts are named `performance_sim.R` (simulation study of Section 3.1), `nca_analysis.R` (pharmacokinetic application of Section 3.2), `volcano_timing.R` (parallelism benchmark of Section 3.3), `dgp_supp_visuals.R` (supplemental visualization of the simulation DGPs), `convert_png_to_jpeg.R` (conversion of PNG figures to JPEG), and `simple_illustration.R` (the single-predictor base R reference implementation of Supplement G).

Supplement A: Multivariate basis example and constraint counting.

Supplement B: Woodbury factorization for structured covariance (full derivation of Section 2.5.2).

Supplement C: REML for correlation parameters.

Supplement D: GLM consistency and asymptotic normality of the constrained estimator.

Supplement E: Simulation DGPs and surface visualizations.

Supplement F: Fixed-penalty RMSE results corresponding to Table 1 with smoothing parameters fixed as described in the supplement.

Supplement G: A single-predictor base R reference implementation of the constrained monomial spline procedure.

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Appendix A: Derivation Using Lagrangian Multipliers

To derive the constrained penalized estimate, solve

$$\tilde{\beta} = \arg \min_{\beta} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|^2 + \frac{1}{2} \beta^\top \mathbf{\Lambda} \beta \right\} \quad \text{subject to} \quad \mathbf{A}^\top \beta = \mathbf{0}. \quad (16)$$

Introducing Lagrange multipliers δ and recalling $\mathbf{G} = (\mathbf{X}^\top \mathbf{X} + \mathbf{\Lambda})^{-1}$, the Lagrangian may be expressed as:

$$\mathcal{L} = -\mathbf{y}^\top \mathbf{X} \beta + \frac{1}{2} \beta^\top \mathbf{G}^{-1} \beta + \delta^\top \mathbf{A}^\top \beta. \quad (17)$$

Setting partial derivatives to zero:

$$\frac{\partial \mathcal{L}}{\partial \beta} = -\mathbf{X}^\top \mathbf{y} + \mathbf{G}^{-1} \beta + \mathbf{A} \delta = \mathbf{0}, \quad (18)$$

$$\frac{\partial \mathcal{L}}{\partial \delta} = \mathbf{A}^\top \beta = \mathbf{0}. \quad (19)$$

From (18), $\beta = \mathbf{G} \mathbf{X}^\top \mathbf{y} - \mathbf{G} \mathbf{A} \delta = \hat{\beta} - \mathbf{G} \mathbf{A} \delta$. Substituting into (19):

$$\mathbf{A}^\top \hat{\beta} - \mathbf{A}^\top \mathbf{G} \mathbf{A} \delta = \mathbf{0} \quad \implies \quad \delta = (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} \mathbf{A}^\top \hat{\beta}. \quad (20)$$

The constrained estimate is therefore

$$\tilde{\beta} = \hat{\beta} - \mathbf{G} \mathbf{A} (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} \mathbf{A}^\top \hat{\beta} = [\mathbf{I} - \mathbf{G} \mathbf{A} (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} \mathbf{A}^\top] \hat{\beta} = \mathbf{U} \hat{\beta}, \quad (21)$$

where $\mathbf{U}$ is the projection defined in (5). For non-zero constraints $\mathbf{A}^\top \beta = \mathbf{c} = \mathbf{A}^\top \beta_0$, the multiplier becomes $\delta = (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} \mathbf{A}^\top (\hat{\beta} - \beta_0)$ and the constrained estimate is $\tilde{\beta} = \mathbf{U} \hat{\beta} + (\mathbf{I} - \mathbf{U}) \beta_0$.

To verify the transformed OLS equivalence (Section 2.5.1), note that

$$\begin{aligned} \mathbf{U} \hat{\beta} &= (\mathbf{I} - \mathbf{G} \mathbf{A} (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} \mathbf{A}^\top) \hat{\beta} \\ &= \mathbf{G}^{1/2} (\mathbf{G}^{-1/2} \hat{\beta} - \mathbf{G}^{1/2} \mathbf{A} (\mathbf{A}^\top \mathbf{G}^{1/2} \mathbf{G}^{1/2} \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{G}^{1/2} \mathbf{G}^{-1/2} \hat{\beta}) \\ &= \mathbf{G}^{1/2} (\mathbf{y}^* - \mathbf{X}^* (\mathbf{X}^{*\top} \mathbf{X}^*)^{-1} \mathbf{X}^{*\top} \mathbf{y}^*) \\ &= \mathbf{G}^{1/2} \mathbf{r}^*, \end{aligned} \quad (22)$$

for $\mathbf{y}^* = \mathbf{G}^{-1/2} \hat{\beta}$ and $\mathbf{X}^* = \mathbf{G}^{1/2} \mathbf{A}$. The constrained solution is thus a left-multiplication of the OLS residual vector $\mathbf{r}^*$ by $\mathbf{G}^{1/2}$.

For non-zero constraints, set $\mathbf{y}^* = \mathbf{G}^{-1/2} \beta_0$ and keep $\mathbf{X}^*$ as above; then $\mathbf{G}^{1/2} \hat{\gamma}$ recovers the $(\mathbf{I} - \mathbf{U}) \beta_0$ component of $\tilde{\beta} = \mathbf{U} \hat{\beta} + (\mathbf{I} - \mathbf{U}) \beta_0$.

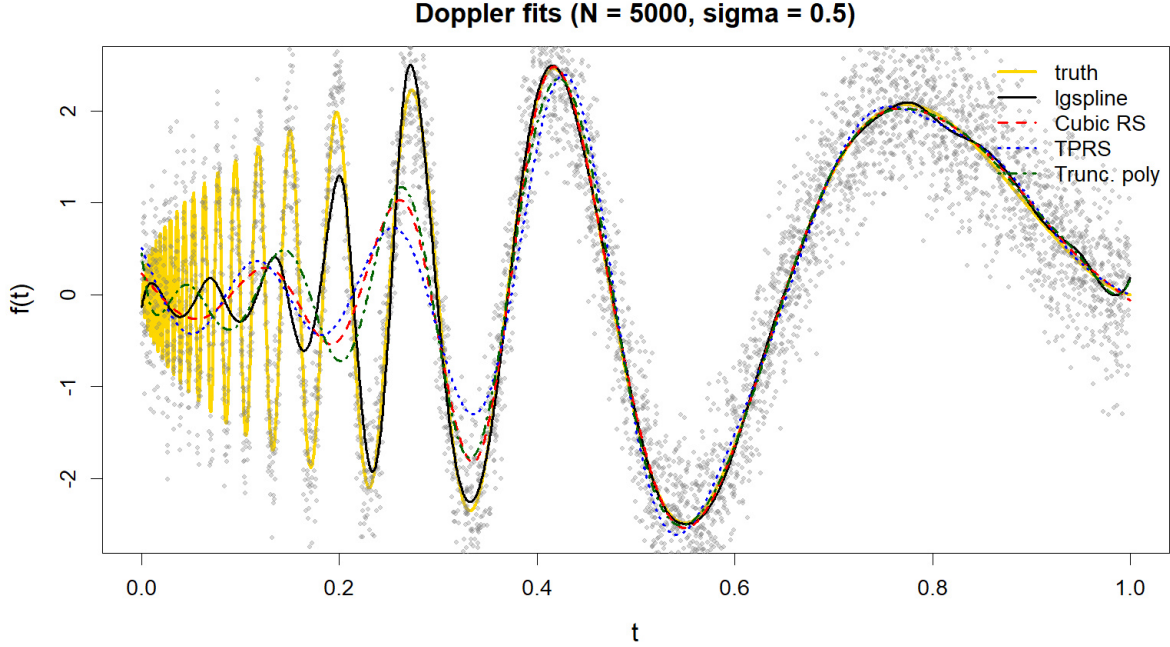


Figure 1: Representative fits on the Donoho-Johnstone Doppler function scaled by 5, at $N = 5,000$ and $\sigma = 0.5$ under automatic smoothness selection. Grey points show the noisy observations ($N = 5,000$); the gold solid line is the true function. The four fitted methods use matching basis dimensions parameterized by $K^* = 16$ knots as described in 3.1; **lgspline** (black solid), Cubic RS (red dashed), TPRS (blue dotted), and the truncated polynomial (dark green dash-dot). All four methods recover the low-frequency structure on $t \in [0.4, 1.0]$, while **lgspline** follows more of the intermediate oscillation than the competing fits. None resolves the first high-frequency oscillation near $t = 0$ at this basis dimension; this is consistent with the lower, but not negligible, **lgspline** RMSE at $q = 1$ reported in Table 1.

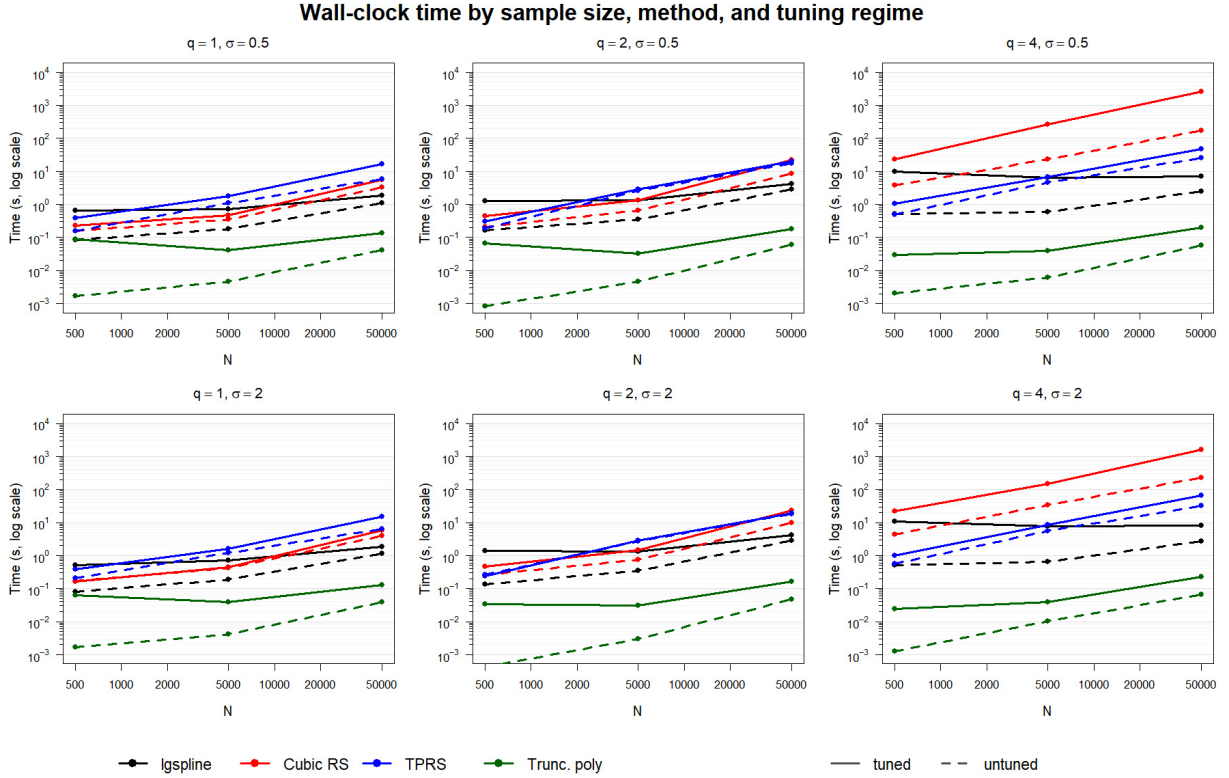


Figure 2: Wall-clock time by sample size for each method, faceted by predictor dimension q and noise level σ . Solid lines show automatic smoothness selection; dashed lines show the fixed penalty. Points are means over $M = 24$ replicates. No wall-clock cap was reached. Exact numerical timings appear in Table 2.

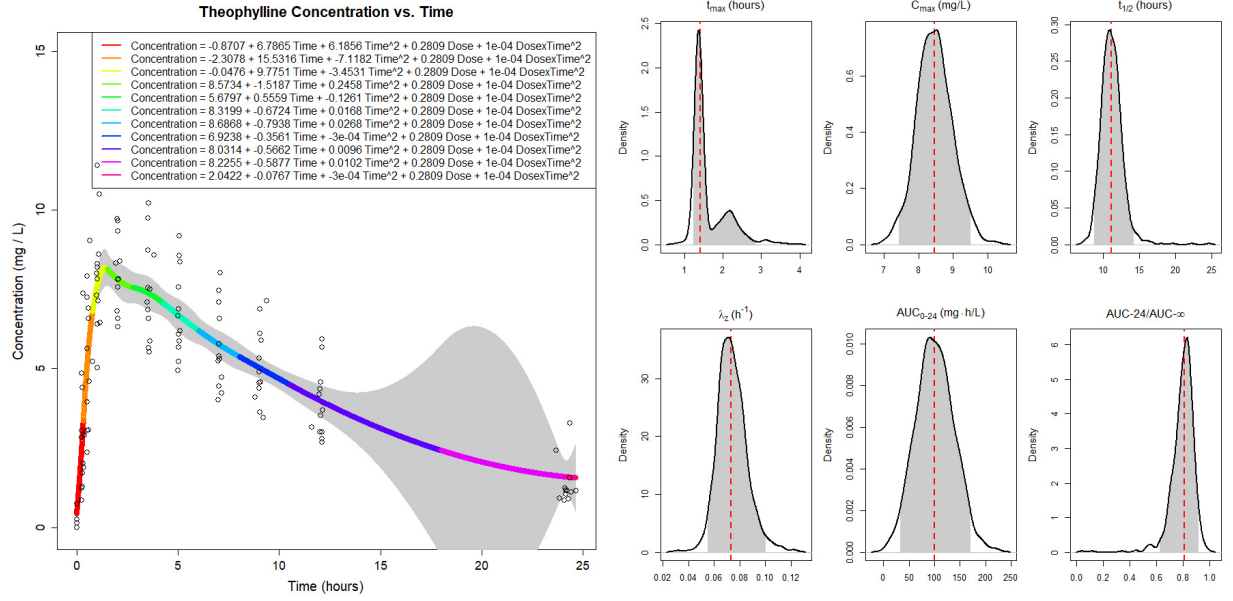


Figure 3: Fitted concentration-time curve at mean dose (left) and posterior densities of the six pharmacokinetic summaries at Dose = 5 mg/kg (right) for the Theophylline analysis. Shape constraints produce a monotone convex terminal-phase decay; the posterior densities carry the fully propagated uncertainty summarized in Table 5.

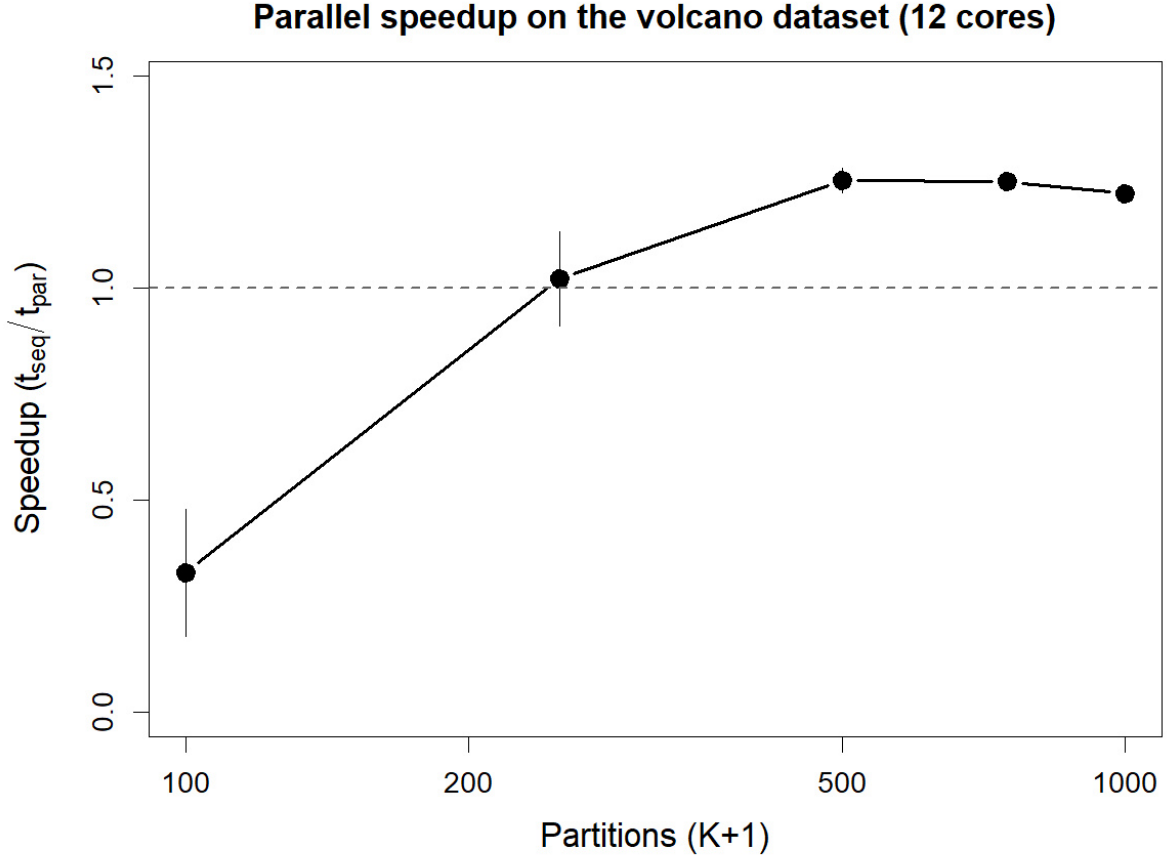


Figure 4: Parallel speedup of `lgspline` on the Maunga Whau volcano dataset as a function of partition count, with 12-core parallelism. Parallelism is costly at $K + 1 = 100$, approximately break-even at $K + 1 = 250$, and reaches only $\sim 1.22\times$ by $K + 1 = 1,000$. The plateau suggests that serial work, including constraint QR operations, remains a limiting part of the current implementation.

Volcano 3-D Map

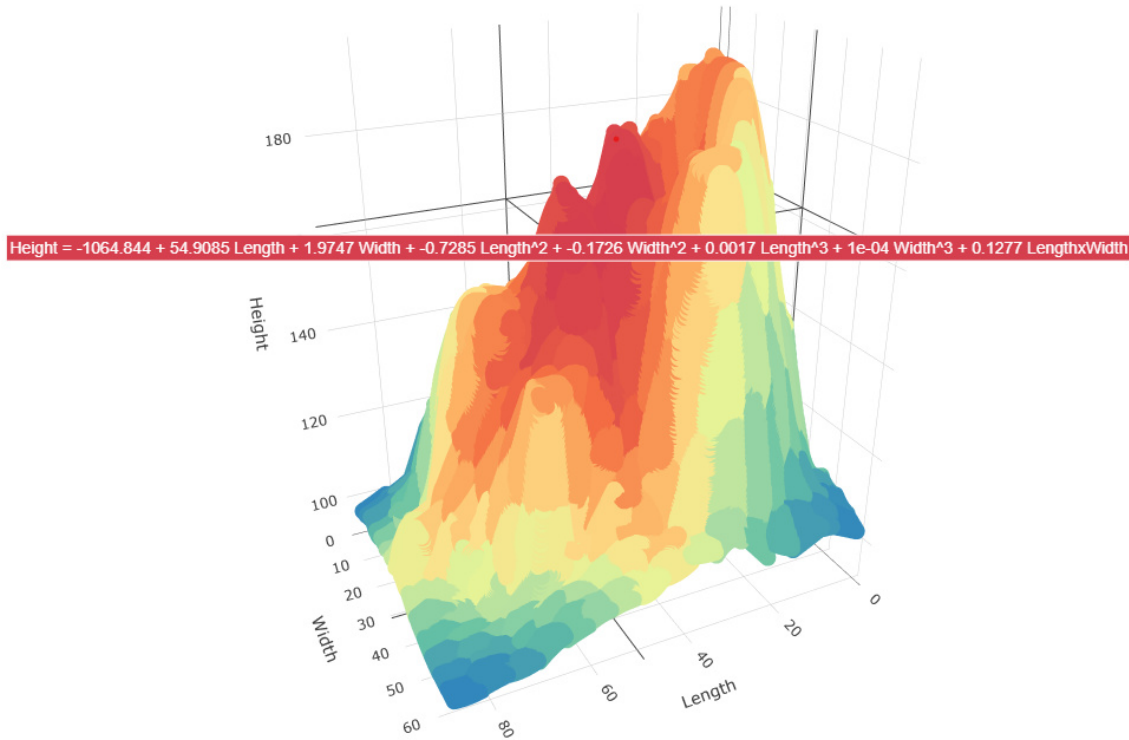


Figure 5: Illustrative `lgspline` fit to the Maunga Whau volcano dataset using a tuned model with $K + 1 = 250$ partitions (colored), fitted-value continuity only, pairwise linear interactions, and cubic expansions for each predictor (Width and Length). The displayed formula is the partition-specific polynomial equation used to generate the fitted surface, accessible by hovering over each partition with a cursor in the interactive `lgspline` output.